

On the characterization and monitoring of building energy demand using statistical process control methodologies



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ABSTRACT

A statistical process control based system and scheme for monitoring, assessment and tracking of energy consumption is presented. The proposed strategy is set up using a Multichannel Structure to model and estimate statistics profile (average and uncertainty) of the energy consumption of a building during a pre-determined cycle, e.g. a week period. These statistics profile is analyzed with statistical process control (SPC) techniques for signaling alarms or reporting interpreted faults of unpredicted or unusual behavior of energy demand. Identification of deviations from predicted or programed consumption profiles modeled by a Multichannel Structure are shown to be easily designed, tuned and implemented with low computation demand of memory and processing speed. Simulated results illustrate the Multichannel Structure programming and parameters tuning. Experimental results are presented to exemplify the proposed strategy.

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1. Introduction

Energy usage in buildings represents a significant percentage of national energy consumption in many countries. In numerical terms, the proportion of energy used in buildings in comparison with the total national consumption is considered to be: 25% for Japan, 28% for China, 37% for European Union, 40% for United States and 42% for Brazil [1–3]. According to Iwaro and Mwasha [4], the building sector is the largest single contributor to worldwide energy consumption since it accounts for 45% of global primary energy resources. This expressive percentage tends to grow in the next years according to the Energy Information Administration, EIA. In their International Energy Outlook, EIA estimated that energy usage in the built environment will grow by 34% in the first half of the XXI century [1]. In this scenario, energy efficiency in facilities is a prime objective of energy policy.

According to the European Union's energy performance of buildings directive, energy consumption is affected by how buildings are designed, built, commissioned, and used [5]. Indeed, to consume energy efficiently it is necessary to build and outfit facilities with materials and systems suited to their location and characteristics. This, however, does not guarantee efficiency. As asserted by the

directive cited above, there is also the matter of how facilities are used. Constructive and technological aspects can only ensure that a particular building is able to consume energy in an efficient manner. To effectively accomplish this goal is imperative to operate with a prudent energy demand target.

Energy efficiency, as an operational action, can only be achieved with a performance-based approach. This is accomplished by collecting, interpreting and reporting energy consumption data. Systems designed to perform these tasks are commonly known as energy-information or energy monitoring and targeting (M&T) systems [6]. The purpose of monitoring and targeting is to relate energy consumption data to weather, production figures, occupational patterns or other measures in such a way that is achieved a better understanding of the energy usage. Once the dynamic of the energy consumption of a building is known it is possible to analyze, compare, set targets and control the energy consumption information.

Energy information systems apply management techniques based on the concept that electricity is a controllable resource that once measured highlights energy saving opportunities. The effectiveness of providing energy information is shown in [7–9]. In these references, energy consumption reporting is used as a tool to increase consciousness and energy savings. Fault detection and diagnosis (FDD) techniques have been used with advantages to detect abnormal energy consumption such as performance degradation, poor maintenance or improper operation of systems [10–15]. Puranik [11], Stuart et al. [12] and Qin and Wang [13]

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α	smoothing factor of the multichannel filter
ε	consumption residuals
g	result of the CuSum test
h	CuSum test threshold
i	channel index
j	cycle index
RMV	Raw Measurement Value
RU	Raw Uncertainty
s	quadratic residuals
$\hat{U}$	estimated uncertainty
ν	CuSum test drift
$V_{(VMV)}$	Validated Measurement Value
$V_{(VU)}$	validated uncertainty
$\hat{X}$	estimated average energy consumption
RMV_{min}	minimal reading value of the ADC of MTS. Used to compute the RMV range
RMV_{max}	maximum reading value of the ADC of MTS. Used to compute the RMV range
$V_{(VMV)_{min}}$	minimal validated measurement. Used to compute the initial uncertainty
$V_{(VMV)_{max}}$	maximum validated measurement. Used to compute the initial uncertainty
m_{av}	average window of the equivalent arithmetic average used to compute the multichannel smoothing factor

use cumulative sum control charts to monitor energy usage and detect abnormalities. Qin and Wang [13] present a hybrid approach utilizing expert rules; performance indexes and EWMA (exponentially weighted moving average) control charts. Seem [10] uses a pattern recognition technique to identify days of the week with similar energy consumption profiles. In a sequel study [10], outlier detection is used to determine if the energy consumption for a particular weekday is significantly different than previous energy consumption. An example of programming and development of an enterprise energy information system are presented in [6]. Swords et al. [6] present a system that integrates data acquisition, analysis and reporting modules and is developed and assessed in two different types of organizations – an industrial site and an education institution.

Energy monitoring, assessment and tracking have been proven to be very useful in enlightening energy consumption and identifying saving opportunities. This practice, however, entails dealing with massive data flow and demands large database capability and specialized professionals capable of identifying and analyzing changes on energy consumption patterns. Both requirements, large database and the need of an expert analysis, still prevents monitoring and targeting in large scales. These are one of the reasons why although commonly endowed with automatic systems, modern buildings often fail to perform energy monitoring and targeting. In most buildings, the collected energy consumption data is simply stored for future offline analysis [16,17]. No further real time processing or analysis is performed. It is, therefore, rare a just in time production of reports on energy usage or proposals on actions likely to promote efficiency.

Systems designed to effectively perform monitoring and targeting should be able to automatically and, continuously measure, analyze and report energy usage just in time. To meet this need, an on-line statistical process control (SPC) based scheme for monitoring and targeting is presented. The strategy is to model and estimate the statistics profile (average and uncertainty) of the energy consumption during a predetermined cycle, e.g. a week period, to monitor, identify and report whenever significant changes occur.

The proposed methodology is based on a low computational cost signal processing algorithm for reducing the amount of data but retaining relevant information. This algorithm uses a similar scheme of a Multichannel Structure [18] to model data into an adaptive structure able to deal with energy consumption inherited peculiarities. SPC metrics and techniques are easily applied to this structure allowing the usage of an adaptive decision process for signaling alarms or reporting interpreted faults. The proposed algorithmic implementation is suitable for real-time analysis and is currently presented as a proof of concept using simulated and playback of historic data trends.

Following it is presented the proposed monitoring and targeting system in Section 2. Section 3 discusses the monitoring and targeting module architecture and setup. Examples illustrating the methodology are given in Section 4. Section 5 presents overall comments and conclusions.

2. Proposed monitoring and targeting system

The proposed Monitoring and Targeting System – MTS is able to automatically and continuously measure, analyze, and report energy usage. The proposed system, illustrated in Fig. 1, comprises two hierarchical levels having the following functionality:

- Plant floor or Lower Level: a data acquisition and processing module as illustrated in Fig. 1 identified as MTS1, MTS2, MTSn. These MTS modules operate as slaves acquiring data from sensors, executing local data validation and compressing data into a Multichannel Structure as presented in the following sections. The MTS modules stores local sensors data and provide under request its own preliminary interpretation of the data it keeps. All MTS slave modules operate autonomously but are coordinated by a supervisory computer via a local network (see Fig. 1).
- Supervisory level or Upper Level: a personal computer running a SCADA system capable of communicating with all the MTS modules via local network. The SCADA application coordinates and concentrates the distributed information in order to keep a just in time plantwide surveillance and to generate an integrated data validation and interpretation of the behavior of the whole building plant.

The proposed system philosophy is based on the self-validating concept proposed by Henry and Clarke in the early 90s [19]. A self-validated instrument is able to detect and correct its own faults and to provide each measurement with an estimate of its uncertainty and standard quality indicators. Similarly, the system proposed is able to provide diagnostic data along with the energy consumption measurement.

Each MTS module is responsible for monitoring the consumption of a building, a sector, a floor or even a room. The module collects raw energy consumption measurements data through a sensor interface, estimates the usual consumption behavior and models it into a Multichannel Structure as shown in Sections 3.1 and 3.2.

The usual consumption behavior or the consumption profile of a building has been discussed in several studies. Different works involving data of short periods from end-use sub metering has shown that while the magnitude of total household energy usage fluctuates during the day, both underlying behavior and end-use consumption are actually highly patterned [20]. Indeed, variables that are strongly determined by human behavior, such as energy or water consumption, usually feature an intrinsic patterned behavior. This behavior can be characterized by daily (day/night), weekly (weekday/weekend) and annual cycles. Daily or weekly profiles are mainly affected by differences on building occupation and external

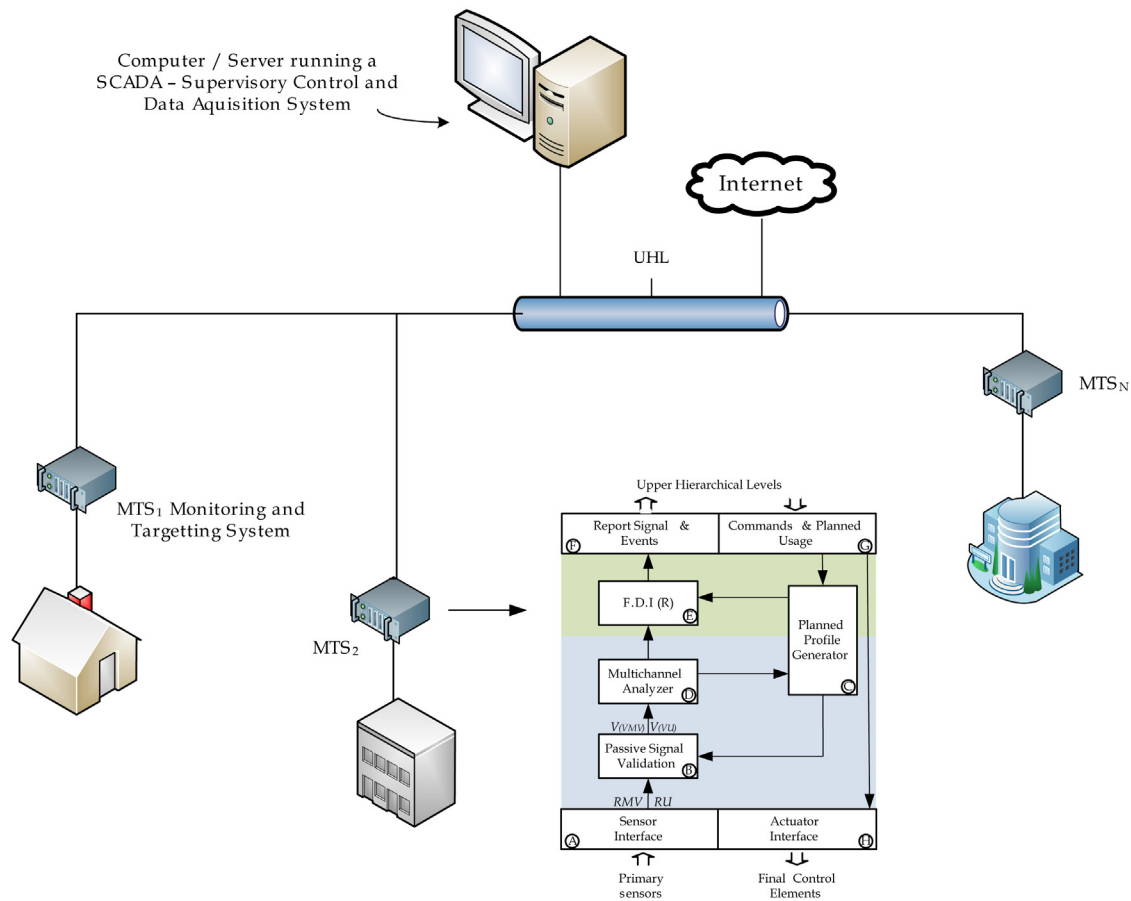


Fig. 1. Block diagram of the monitoring and targeting system from a plantwide perspective.

weather conditions, while annual profiles are mostly influenced by seasons, with different requirements for lighting, heating and cooling.

According to Lutzenhiser [20] daily and weekly profiles are pretty stable over time, even though they often differ significantly from one another in their energy usage patterns. Martinez-Ortiz et al. [21] claim that the regular behavior of a building can be categorized in accordance to the kind of activities performed in the building. Martinez-Ortiz et al. [21] present profiles for four different buildings types shown in Fig. 2. According to the author, office buildings (Fig. 2a) tend to have higher consumption during officer hours and a reduced consumption during nights and weekends. While hotels (Fig. 2b) do not show substantial differences in consumption between weekdays and weekends. University buildings (Fig. 2c) usually display two different patterns of behavior during weekdays. The different patterns derive from the typical school schedule. Industrial buildings (Fig. 2d), on the other hand, usually do not present a clear weekly pattern of behavior.

The proposed Multichannel Structure (see Section 3.1) compresses the consumption data according to its statistical profile. An example of how this data compression is done is shown in Fig. 3. Fig. 3a shows a typical energy consumption time series during 3 months and Fig. 3b shows the statistical profile of the consumption in a weekly period. Note that Fig. 3b represents the weekly average energy consumption and its 3-sigma variability.

By compressing information, the Multichannel Structure dramatically reduce the amount of data storage and data processing demands. The Multichannel Structure facilitates the usage of an automated statistical decision process for signaling alarms and reporting interpreted faults of unpredicted or unusual behavior of

current energy demand of a building. Identification of deviations from predicted or programed consumption profiles modeled by the Multichannel Structure are straightforwardly designed, tuned and implemented with low cost microcontroller systems with limited computation resources (memory and processing speed) [25]. Detected faults are then reported to the supervisory level.

The supervisory system is responsible for data storage, integration and mapping. The supervisory system is able to aggregate consumption of various environments in larger ones or consumption units created virtually on the server. In a building provided with MTS modules in every one of its rooms, the energy consumption by floor is composed of the sum of several MTS modules. This distributed monitoring strategy allows a modular tracking of faults.

3. Monitoring and targeting modules

The monitoring and targeting strategy executed in each MTS module is illustrated in Fig. 4a and b is shown its data structure.

The monitoring and targeting strategy initiates at collecting raw measurement data from sensor devices through the Sensor Interface (indicated as A in Fig. 4a). The collected data is passively validated in the Passive Signal Validation block (indicated as B in Fig. 4a) as shown in Section 3.1. The Multichannel Analyzer block (indicated as C in Fig. 4a) packs the validated variables into Multichannel Structures as shown in Section 3.2. These structures are passed through the Fault Detection, Identification and Reporting block (indicated as E in Fig. 4a). In this block, data is evaluated using statistical process control tools. Whenever necessary, alarms and messages are passed to Upper Hierarchical Levels (UHL) through

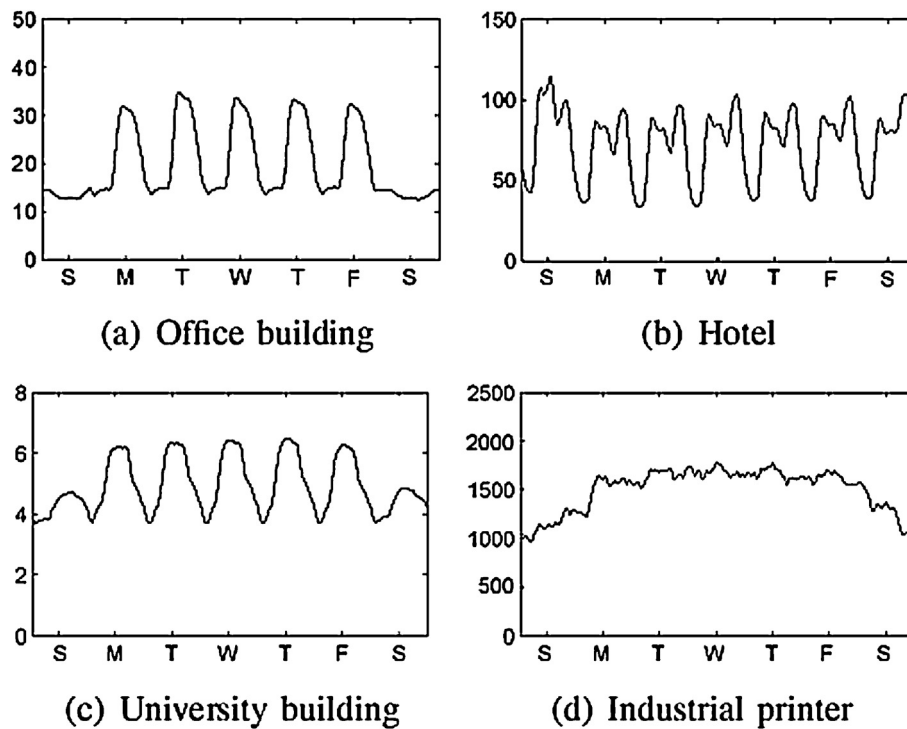


Fig. 2. Typical electricity weekly profile for different types of buildings. From Martinez-Ortiz et al. [21].

the Reports Signals And Events interface (indicated as F in Fig. 4a). From the Commands and Planned Usage interface (indicated as G in Fig. 4a) are sent information on the scheduled activities that are shaped into a Planned Profile by the Planned Profile Generator (indicated as C in Fig. 4a) as seen in Section 3.3. This information is used to filter alarms in the Fault Detection, Identification and Reporting block and to estimate the measurand case no data is received from the sensor interface. UHL also sends commands to actuated control elements through the Actuator Interface (indicated as H in Fig. 4a).

The MTS object is composed by the variables shown in Fig. 4b. Two variables assigned to each sensor: the actual measurand, denominated Raw Measurement Value (RMV), and its Raw Uncertainty (RU) estimated offline from sensor calibration. These variables, after being passive validated are called Validated Measured Value ($V_{(VMV)}$) and Validated Uncertainty ($V_{(VU)}$). Two pairs of Multichannel Structures are estimated by the Multichannel Analyzer:

a short length average and uncertainty and a long length average and uncertainty. These structures are used by the Fault Detection, Identification and Reporting block to generate alarms and diagnostics. In addition to these, the MTS object is also composed of auxiliary variables for computing the Multichannel Structures, as shown in Sections 3.1 and 3.2 and for the statistical analysis shown in Section 3.6.

The monitoring and targeting strategy can be divided into the following six steps to be described in the following sections:

- 1 Data prefiltering and sorting into channels
- 2 Current and historical models updating
- 3 Planned profile generator
- 4 Data labeling and device status
- 5 Setup of the MTS module
- 6 Fault detection

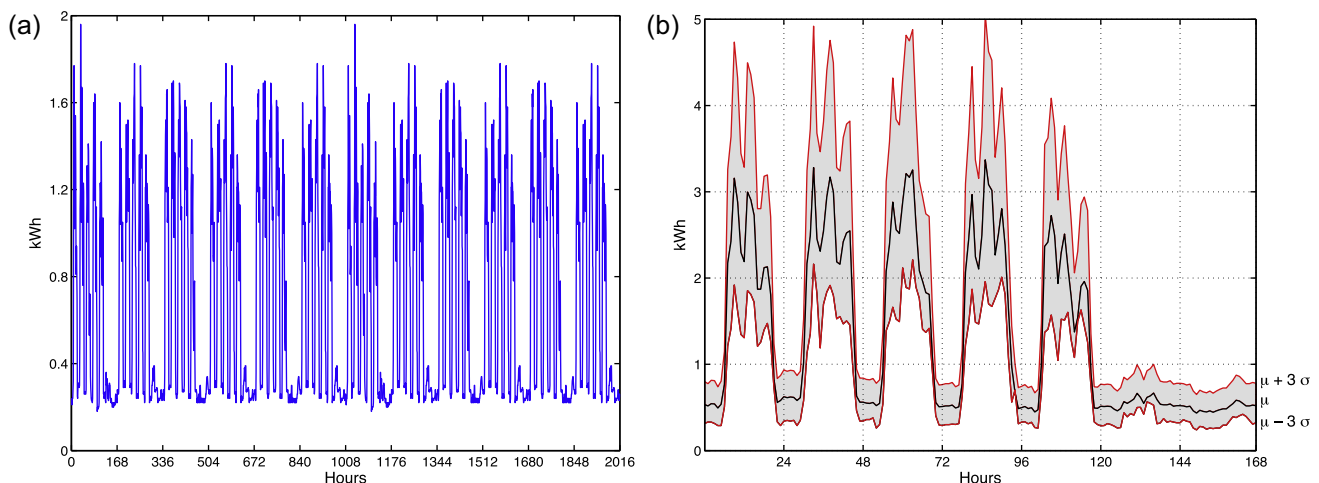


Fig. 3. Energy consumption time series and its statistical behavior.

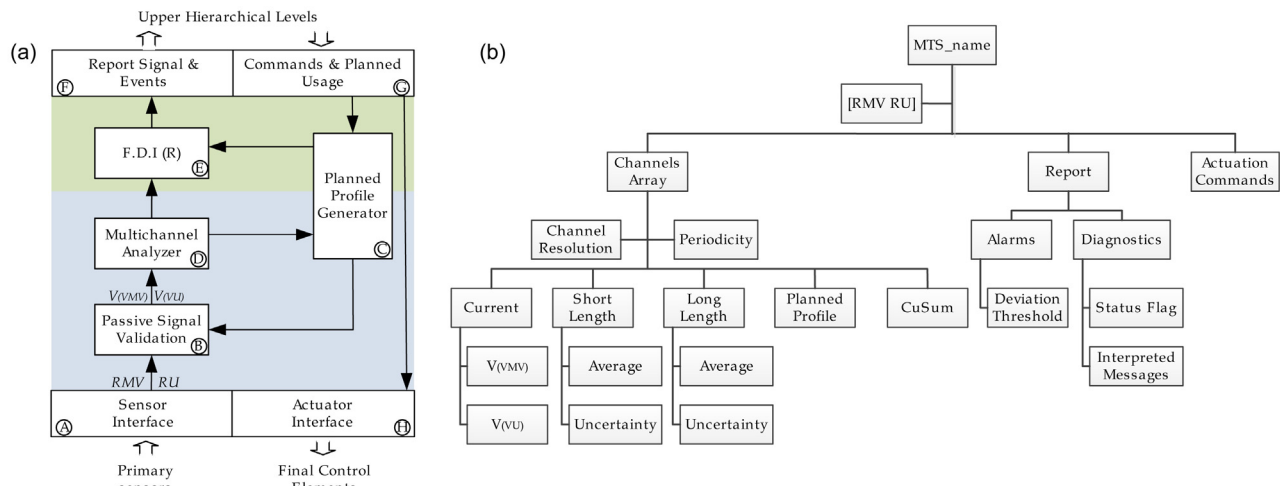


Fig. 4. Monitoring and targeting strategy and its data structure.

3.1. Data prefiltering and sorting into channels

Raw measurand is collected from sensor devices (at adjustable intervals) through the Sensor Interface. As stated previously, two variables are assigned to each sensor: the actual measurand, denominated Raw Measurement Value (RMV), and its Raw Uncertainty (RU) estimated offline from sensor calibration.

Both variables are passed along to the Passive Signal Validation interface. This interface initially checks RMV against spike and outliers. Then, if necessary, an average filter is used to modify the RMV sampling interval according to the desired resolution for the analysis. The median filter is used when, for example, the raw data is collected with a sampling interval of 1 min but it is understood that a longer time interval is sufficient for the analysis. Sampling intervals less than 15 min are usually considered unnecessary since this is the typical utility billing interval. After this first data processing stage, the consumption data is now denominated Validated Measured Value ($V_{(VMV)}$) and the RU increased with the uncertainty of the filtering processes is denominated Validated Uncertainty ($V_{(VU)}$). The Validated Measured Value and the Validated Uncertainty are then passed along to the Multichannel Analyzer block (indicated as C in Fig. 4a). This interface models the energy consumption, $V_{(VMV)}$, according to its statistical behavior into Multichannel Structures.

A Multichannel Structure is composed of a number of channels placed one beside the other as shown in Fig. 5. The number of channels contained in the structure depends on two variables: the channel resolution and the periodicity of the structure. The resolution is given by the amount of time constrained in each channel and is limited by the sampling interval of the measurand. The periodicity represents the underlined cyclical behavior of the building and can be of days, weeks, months or years. Fig. 5 represents a

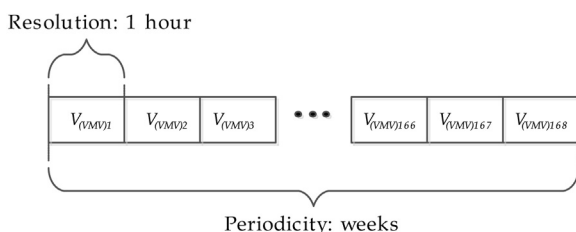


Fig. 5. Multichannel Structure sized for a week pattern with 1-h resolution and 168 channels.

Multichannel Structure used to model the energy consumption of a building with weekly patterned usage and channel resolution of 1 h. The structure is therefore constituted by 168 channels, each one representing one of the 168 h of a week. Each channel is assigned with the amount of energy consumption occurred during an hour. The first channel represents the consumption occurred on Monday from 00:00 to 00:59 and the last channel represents the consumption occurred on Sunday from 23:00 to 23:59.

The Multichannel Structure illustrated in Fig. 5 (periodicity of weeks and channel resolution of 1 h) is used to model the first 5 weeks of the data shown in Fig. 3a. Fig. 6a illustrates how the data is folded into the multichannel array. Fig. 6b shows the trend values of the energy consumed in channel number 56 (8 a.m. of Wednesday) throughout the weeks.

To model the average energy consumption and its uncertainty, a pair of Multichannel Structures are used: one to model the average consumption of a channel and another to model its uncertainty. Initially there is only one value representing the consumption of each channel. However along subsequent weeks, when new measurements become available, it is possible to estimate the values of each channel as shown in Fig. 7. It is important to note that the statistical analysis (estimation of average and uncertainty) is performed intra channels. That is, the consumption measured, for example $V_{(VMV)12}$, on Monday between 1:00 and 1:59 is compared only with the ones measured on Mondays in the same time interval, e.g. $\{ \dots, V_{(VMV)22}, V_{(VMV)32}, \dots, V_{(VMV)n2} \}$.

The Multichannel Analyzer (Fig. 8) generates two models that are continuously updated and stored: a Short Length Model and a Long Length Model. Each model is compound with two multichannel arrays one average and other for the uncertainty or deviation from the mean. The Long Length Model represents the average behavior for a longer past period, e.g. 20 weeks. The Short Length Model is more frequently updated, e.g. 2 last weeks average model.

3.2. Current and historical models updating

At each new cycle (e.g. a week in Fig. 7), average energy consumption ($\bar{X}$) is computed for every channel according to Eq. (1).

$$\hat{\bar{X}}_{j,i} = \hat{\bar{X}}_{j,i} + \alpha [V_{(VMV)_{j,i}} - \hat{\bar{X}}_{j,i}] \quad (1)$$

Where i is the channel index, j is the cycle index and α is the smoothing factor ($0 < \alpha < 1$).

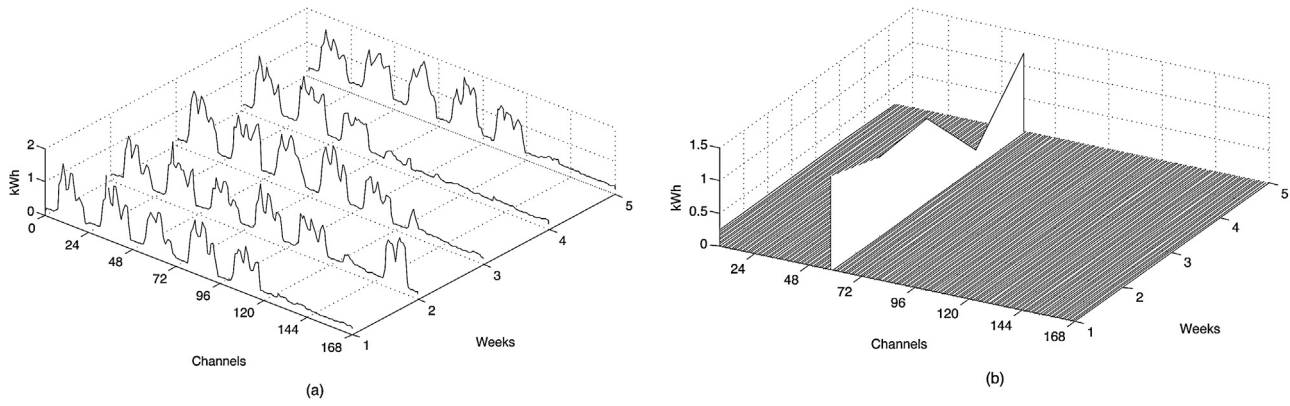


Fig. 6. 3D visualization of data folding in a Multichannel Structure. (a) Data trend folded in week periods and (b) a channel data trend along the past weeks.

The estimated uncertainty (U) of the measured values is computed as the maximum value among validated uncertainty and the estimated sample variance according to Eq. (2).

$$\hat{U}_{j,i} = \max(\hat{\sigma}_{j,i}, V_{(VU)_{j,i}}) \quad (2)$$

The estimated variance is computed to Eq. (3).

$$\hat{\sigma}_{j,i}^2 = \frac{2-\alpha}{2(1-\alpha)} \left[\hat{X}_{j,i}^2 - (\hat{X}_{j,i})^2 \right] \quad (3)$$

where

$$\hat{X}_{j,i}^2 = \bar{X}_{j,i}^2 + \alpha \left[V_{(VMV)_{j,i}}^2 - \bar{X}_{j,i}^2 \right]$$

Two different smoothing factors are used to generate a Short a Long Length Models. The parameter α relates to the arithmetic mean as $\alpha = 2/(m_{av} + 1)$, where m_{av} is the number of data points. The Long Length Model, that represents the average behavior for a longer past period, can be computed, e.g. using an m_{av} of 20 weeks. While the Short Length Model that holds and weighs the

more recent data, is computed using an m_{av} of 1 (no filtering) or 2 cycles (e.g. weeks) in energy profiling.

3.3. Planned profile generator

If a fault occurs and no measurand is available, the Validated Measured Value and Validated Uncertainty is estimated according to the usual or planned occupational usage in the Planned Profile Generator (indicated as C in Fig. 4a). The planned occupational use, denominated Planned Profile, consists of a percentage estimate of the usual consumption scheduled to occur in a specific time. Fig. 9 shows the Planned Profile for a week-period in three different scenarios. The first one (Fig. 9 – week 1) represents a typical week, when no change is planned to occur. During typical periods, Planned Profile value is equal to 1 during working hours. The second scenario illustrated (Fig. 9 – week 2) outlines a period in which it is known that the building usage is going to be increased during the working days (Monday to half-day Wednesday due to a Holiday). In this case, the Planned Profile is equal to 2 during working hours and is equal to 0 during the holiday. The last week

	Monday 0:00 to 0:59	Monday 1:00 to 1:59	Monday 2:00 to 2:59		Sunday 21:00 to 21:59	Sunday 22:00 to 22:59	Sunday 23:00 to 23:59
Week 1	$V_{(VMV)11}$	$V_{(VMV)12}$	$V_{(VMV)13}$		$V_{(VMV)1(i-2)}$	$V_{(VMV)1(i-1)}$	$V_{(VMV)1i}$
Week 2	$V_{(VMV)21}$	$V_{(VMV)22}$	$V_{(VMV)23}$	...	$V_{(VMV)2(i-2)}$	$V_{(VMV)2(i-1)}$	$V_{(VMV)2i}$
Week n	$V_{(VMV)n1}$	$V_{(VMV)n2}$	$V_{(VMV)n3}$		$V_{(VMV)n(i-2)}$	$V_{(VMV)n(i-1)}$	$V_{(VMV)ni}$
Average	$\hat{X}_{n1}$	$\hat{X}_{n2}$	$\hat{X}_{n3}$	...	$\hat{X}_{n(i-2)}$	$\hat{X}_{n(i-1)}$	$\hat{X}_{ni}$
Uncertainty	$\hat{U}_{n1}$	$\hat{U}_{n2}$	$\hat{U}_{n3}$	...	$\hat{U}_{n(i-2)}$	$\hat{U}_{n(i-1)}$	$\hat{U}_{ni}$

Fig. 7. Multichannel array structure and data folding by a week period.

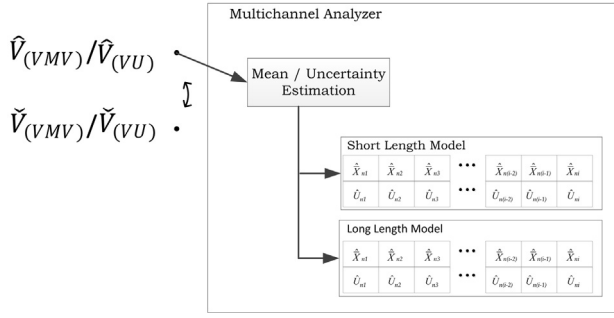


Fig. 8. Multichannel analyzer interface.

(Fig. 9 – week 3) outlines a scenario in which the services will be reduced in a scale of 50% (due to a scheduled training). In this case the Planned Profile is equal to 0.5 during the working hours.

If the Planned Profile is available $V_{(VMV)}$ and $V_{(VU)}$ can be estimated by the Planned Profile Generator as shown in Fig. 10. $V_{(VMV)}$ is computed to be a percentage of the known long length (or historical) consumption of the building, i.e., it is the multiplication of the Usage Function and the Long Length Average Consumption, while $V_{(VU)}$ is initially estimated to be the Historical Uncertainty. However $V_{(VU)}$ value is increased with time indicating the reduced confidence in the value.

3.4. Data labeling and device status

It is interesting to provide the information of weather the $V_{(VMV)}$ and $V_{(VU)}$ values are measured or estimated. Therefore a status flag qualifier is generated by the system. Five measurement status flag are provided:

- 1 *Secure*: indicate that raw data is collected from a sensor in good condition and no fault is detected. $V_{(VMV)}$ is calculated normally.
- 2 *Blurred*: raw data is still being obtained but a fault condition is detected. $V_{(VMV)}$ is projected from past history. $V_{(VU)}$ is increased with time indicating the reduced confidence in this projected $V_{(VMV)}$. This state is temporary.

- 3 *Dazzled*: this stage is an evolution of the blurred stage. If data is considered blurred for more than an specific number of channels, status flag is changed to dazzled, and an alarm is set.
- 4 *Planned*: raw data is known to be uncorrelated with the true process variable. Long Length and Short Length Models are computed using an lower smoothing factor.
- 5 *Blind*: raw data is not available and it is not possible to estimate $V_{(VMV)}$, or it has been estimated for more than a specific number of cycles. This number can vary from facility to facility.

3.5. Setup of the MTS module

The MTS parameters for a new building are set as follows:

- 1 Periodicity expected for the monitored variable e.g. day, week or month.
- 2 Channel resolution (e.g. 15 min is a common choice for Power meters).
- 3 Raw Measurement Value range $[RMV_{min} \ RMV_{max}]$, e.g. ADC (Analog-to-Digital Converter) scale.
- 4 Raw Measurement Uncertainty that is set to a minimum as the sensor calibration uncertainty or ADC resolution.
- 5 Validated Measurement Value range $[V_{(VMV)_{min}} \ V_{(VMV)_{max}}]$, e.g. a Power meter with 10 kW range is $[0 \ 10]$ with Unit = 'kW'.
- 6 Initial uncertainty of the Multichannel model. Without previous information a full-scale variance is assumed, i.e. $(V_{(VMV)_{max}} - V_{(VMV)_{min}})^2$.
- 7 Average window, m_{av} . The multichannel filter time constant is specified considering an exponential average window equivalent to an arithmetic average, i.e. $\alpha = 2/(m_{av} - 1)$. The default value for m_{av} is 3 which implies that 64% of the channel corresponds to the measurements of the last 2 cycles. The uncertainty of the estimated Multichannel model approaches the measured uncertainty with this same speed.

An example of the parameters used to setup the ADC interface of the MTS module is shown in Table 1. In this example the ADC interface of the MTS module is set with a weekly periodicity and channel resolution of 15 min. The initial Reference Model is given by the designed power demand modulated by (i.e. multiplied

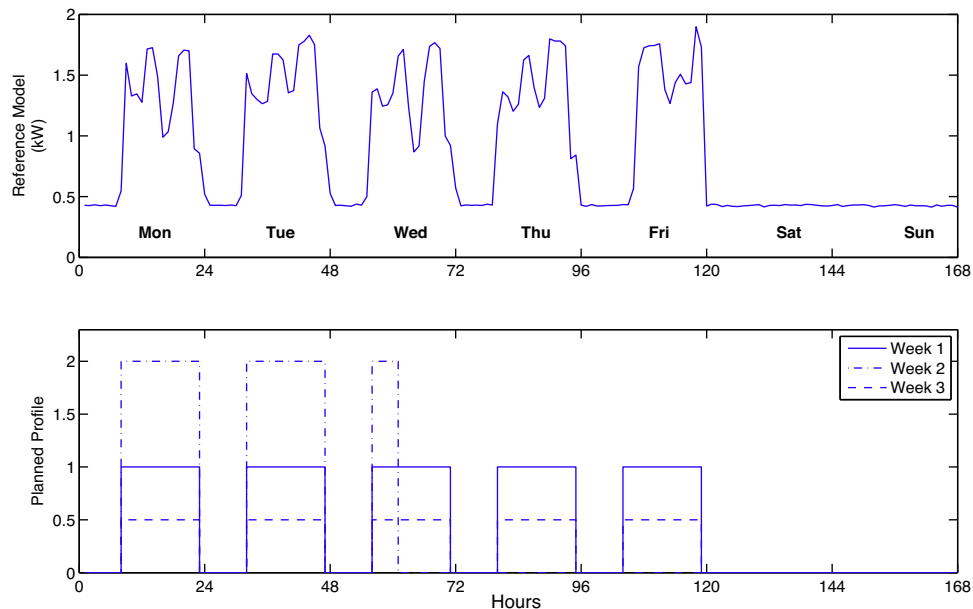


Fig. 9. Planned profile.

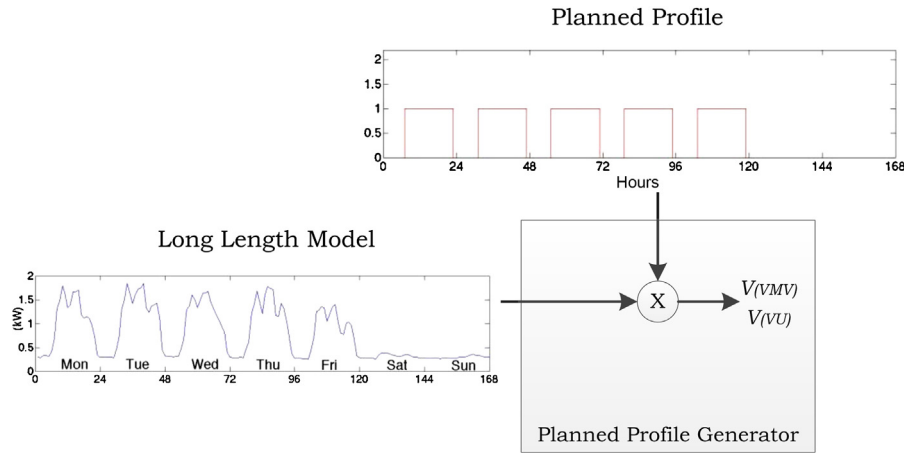


Fig. 10. Planned profile generator.

Table 1
Example of a MTS ADC setup parameters.

Parameter	Value
ADC bits	10
Periodicity	1 week
Resolution	15 min
m_{av}	3
RMV_{max}	1024 ^a
RMV_{min}	0
RU	0.002 ^a
$V_{(VMV)_{max}}$	10
$V_{(VMV)_{min}}$	0
Initial variance	100
α	0.5

^a $2^{ADCbits}$.

^a $2(0.5^{ADCbits})$.

by) the channel profile. The Reference Model uncertainty is set to a maximum e.g. full scale. This way all alarms are silenced until the estimated uncertainty settles down close to its actual value. The channel memory of past values is designed to adapt to seemingly changes. The design parameter, m_{av} , is an equivalent average window of an exponential filter with a time constant, in terms of number of samples, equals to $(m_{av} + 1)/2$.

Fig. 11 illustrates the changes in the uncertainty values over the cycles (in this case, weeks). Initially the uncertainty is maximum and as the cycles pass by it converges to the real value. Note that since the average window is set to a default value, i.e. 3, it requires 10 weeks for the uncertainty to converge. This period is consistent with the expected 4–5 time constants.

3.6. Fault detection

Once the Multichannel Structures are computed, they are passed along, in conjunction with the status flag information, to the Fault Detection Identification and Reporting block (indicated as E in Fig. 4a). In this block, statistical process control (SPC) tools are used to monitor and evaluate consumption.

For each channel the residuals among the measured value ($V_{(VMV)}$) and the computed short length average values are computed. In a highly automated building the residuals would be close to zero. In other building types residues have larger values. In either way, as long as the consumption has a patterned behavior, if no change occurs, residues will resemble a white noise. When ever a change occurs, the residuals become ‘larger’ in some sense (either the mean, variance or both changes). Therefore a change detection

algorithm can be used to locate substantial changes and to generate alarms and reports.

Several algorithms for fault detection can be used on the Fault Detection Identification and Reporting interface. Currently is employed the cumulative sum (CuSum) test of Page [22]. The CuSum analyzes each channel residuals, i.e. deviations from each channel measured value ($V_{(VMV)}$) over its current estimated average (Eq. (4)).

$$\epsilon_{j,i} = V_{(VMV)_{j,i}} - \hat{X}_{j,i} \quad (4)$$

The CuSum statistical test $g_{j,i}$ is computed for each channel i at each new cycle j (Eq. (5)). Since the CuSum is computed intra channels it is called Multichannel CuSum. The test sums its own value to the measured residuals, hence the name “cumulative sums”. If the calculated value exceeds a threshold, h , it is given an alarm, as shown in Eq. (5).

$$s_{j,i} = (\epsilon_{j,i})^2$$
$$g_{j,i} = \max(g_{j-1,i} + s_{j,i} - \nu, 0) \quad (5)$$

$$\text{Alarm} \quad \text{if} \quad g_{j,i} > h \quad (6)$$

If no changes occur, the test statistics will drift away similar to a random walk. To prevent false alarms whenever several statistical tests show a consecutive positive value, a small drift term ν is subtracted from the test for each iteration. The threshold, h , and drift, ν , values are design parameters of the test and are different for each building.

Once an alarm is generated by the CuSum test, a diagnosis is done taking in consideration other parameters such as the current status flag. Joint analysis of CuSum alarms with status flag information can determine the suppression of alarms or generation of reports. For example, if the CuSum test indicates an alarm situation but the status flag indicates that this was planned, the alarm can be suppressed. Important information or reports are sent to the supervisory level.

4. Example of the monitoring and targeting strategy

In order to exemplify the usage of the proposed methodology, two examples are presented. The first one is a case study performed using consumption data simulated by EnergyPlus [23]. EnergyPlus is an energy analysis and thermal load simulation program based on users’ description of a building from the perspective of the building’s physical make-up, associated mechanical systems, etc. The

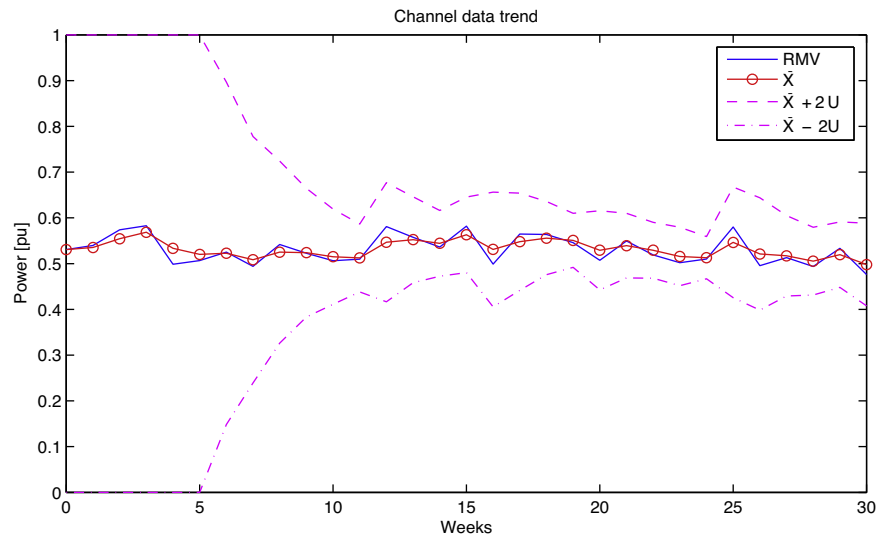


Fig. 11. Evolution of the mean and uncertainty values for one channel.

second example presents real data from an educational building in Brazil [24].

4.1. Simulated data

The 'based case' simulated in the program consists of a small house (Fig. 12) equipped with a variable air volume air conditioning system.

Data are simulated with the sampling average of 15 min during the period of three months. A weekly occupational profile is simulated and a channel resolution of 1 h is considered ideal. Therefore the Multichannel Structure used to model the consumption consists of 168 channels each one representing the 168 h of a week.

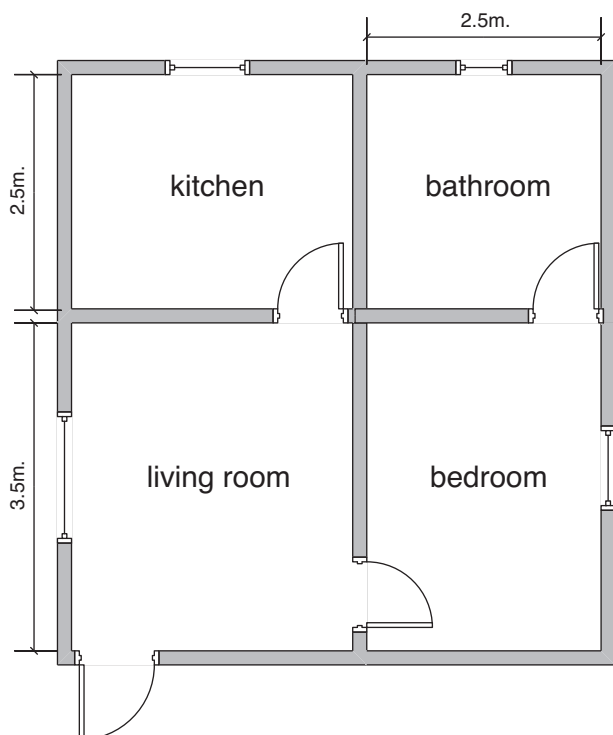


Fig. 12. Plan of the case-study house.

In the first week the average consumption is the actual energy consumption and the uncertainty is $V_{(VU)}$ (sensor uncertainty increased with the uncertainty of the passive validation). Fig. 13 shows the Short Length Average Multichannel Structure, the Short Length Uncertainty Multichannel Structure and the Status flag for the first week.

On subsequent weeks short and long length historical average and uncertainty are computed for each channel. On week 5 (Fig. 14), at 8 a.m. on Monday a spike occurs. Immediately the status flag changes to Blurred, the spike is removed from data and the uncertainty increases. Later this week a nonusual event (no occupancy) is expected to occur. During the period of time that this event is scheduled, the smoothing factor is attenuated in order to minimize the importance of this unusual consumption in the computation of the Multichannel Structures. Status flag changes to Planned.

Note that since this unusual consumption was planned it does not influence the average and uncertainty values. Since the unusual consumption was planned, no alarm is generated.

If no information was previously known (that is if it was not stamped as planned) average and uncertainty values would be affected as shown in Fig. 15.

On week 8 communication lost with sensors is emulated. Fig. 16 shows that when communication is lost the status flag changes to Blind. During the first 2 h (channels) when no data is received, $V_{(VMV)}$ and $V_{(VU)}$ are estimated in accordance with the Usage Function and with the Historical Average. After 2 h (this time is used-defined), $V_{(VMV)}$ is set to zero and uncertainty increases with time.

For each channel a CuSum test is performed in order to detect abnormal consumption. Fig. 17 shows the CuSum Test Statistics and the generated alarms for the week 8 of monitoring. An alarm is generated as soon as communication is lost with the sensor. Note that the alarm information indicated that during the first two channels although no data is being received, $V_{(VMV)}$ and $V_{(VU)}$ are estimated from long length (historical) values. After this, the alarm information indicates only that communication is lost.

4.2. Real data

The facility is part of the educational complex of a Brazilian University Campus. The building, built among the years of 2000 and 2001, is comprised of classrooms and laboratories and presents a weekly occupancy profile. Classes take place in all three periods.

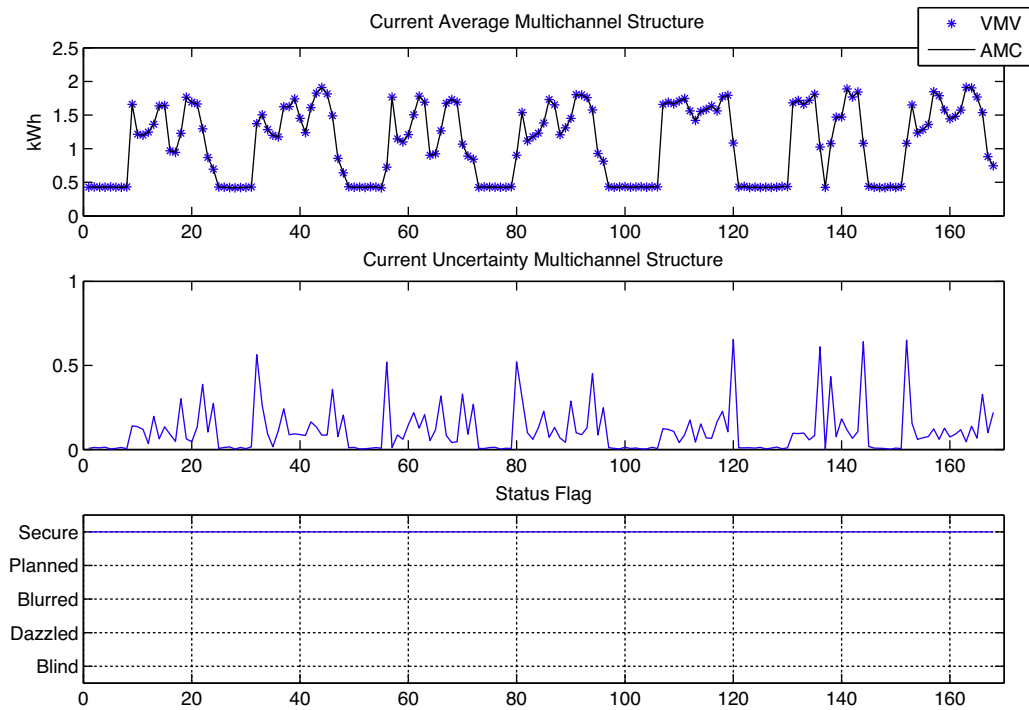


Fig. 13. Short length multichannels and status flag – week 1 of simulated data.

The morning shift starts at 07:30 and ends 12:50. In the afternoon, classes take place from 13:00 to 18:40. And during the night shift classes start 19:00 and end 22:30. The installation has 3200 m² and about 30% and 40%, respectively, of its load installed in the lighting and ventilation systems.

The building's electricity consumption data from a week-long measurement is presented in Fig. 18. A simple visual inspection of

the data indicates that each week day presents a particular consumption profile. A more detailed examination confirmed the data to have a weekly consumption pattern.

Electricity energy consumption was measured and recorded during eight weeks. Fig. 19 shows the Validated Measurement Values plotted in weekly periods (presented in three weeks for each graph).

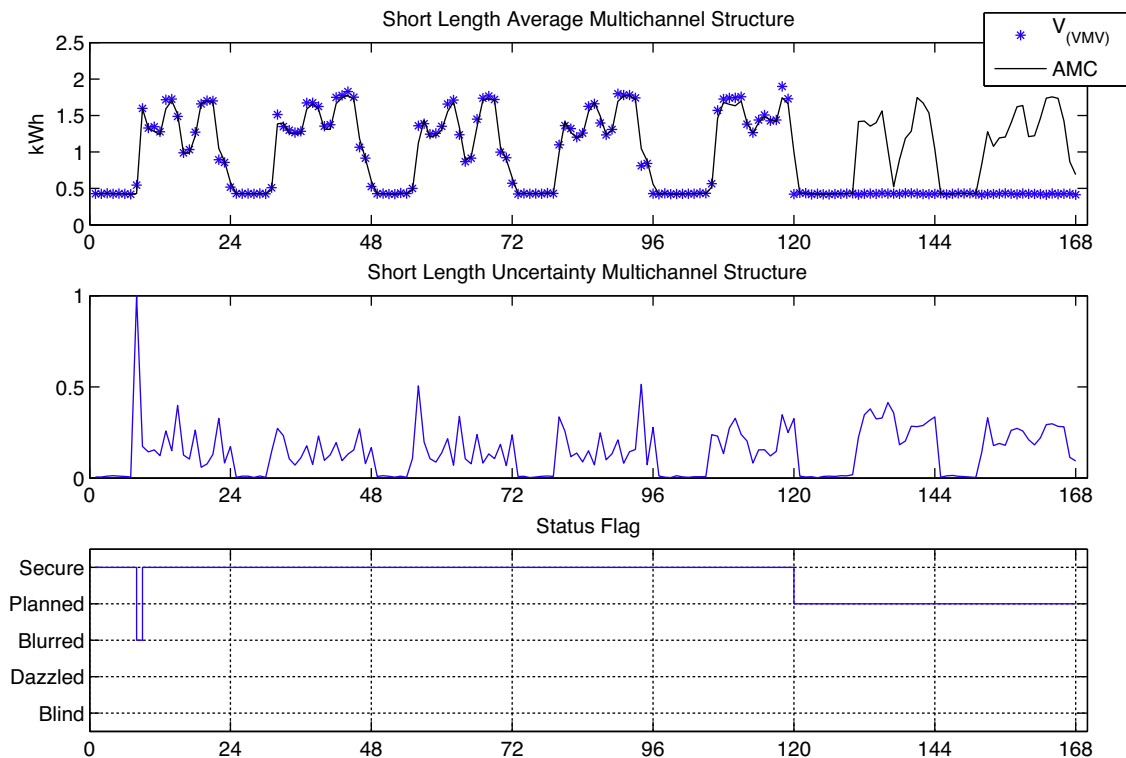


Fig. 14. Short length multichannels and status flag – week 5 of simulated data.

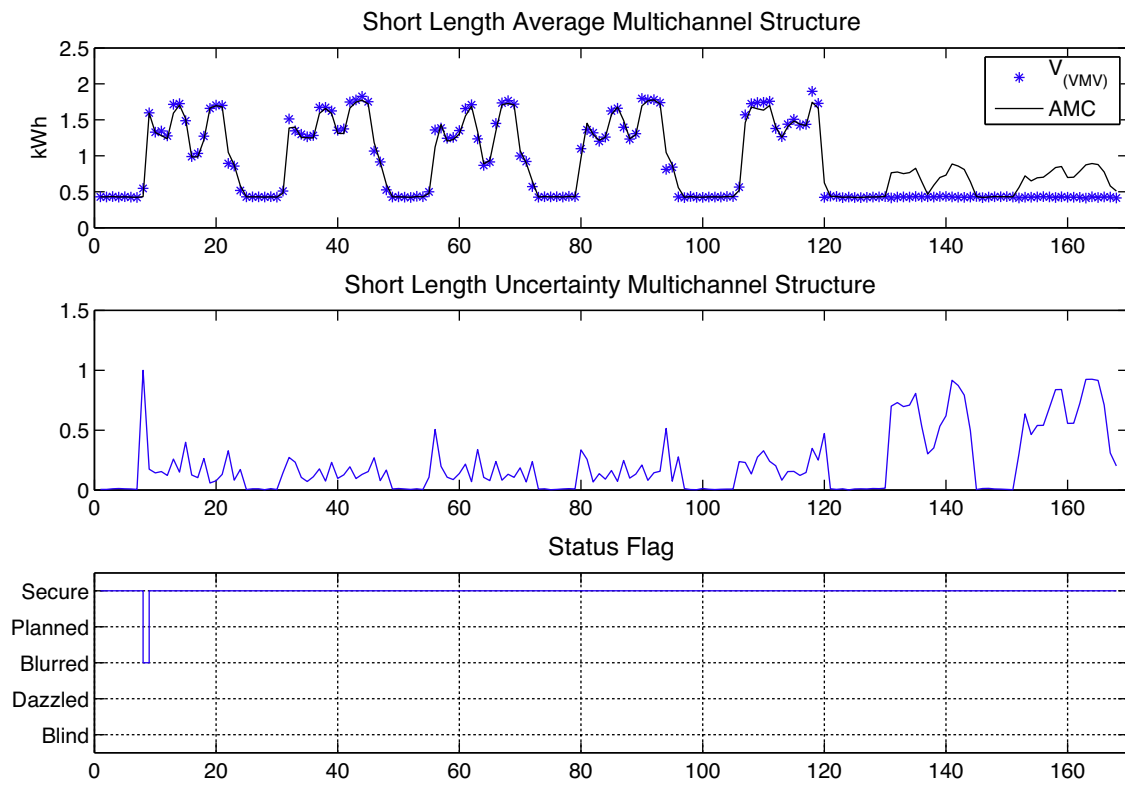


Fig. 15. Short length multichannels and status flag – week 5 of simulated data. Case no planned event is scheduled.

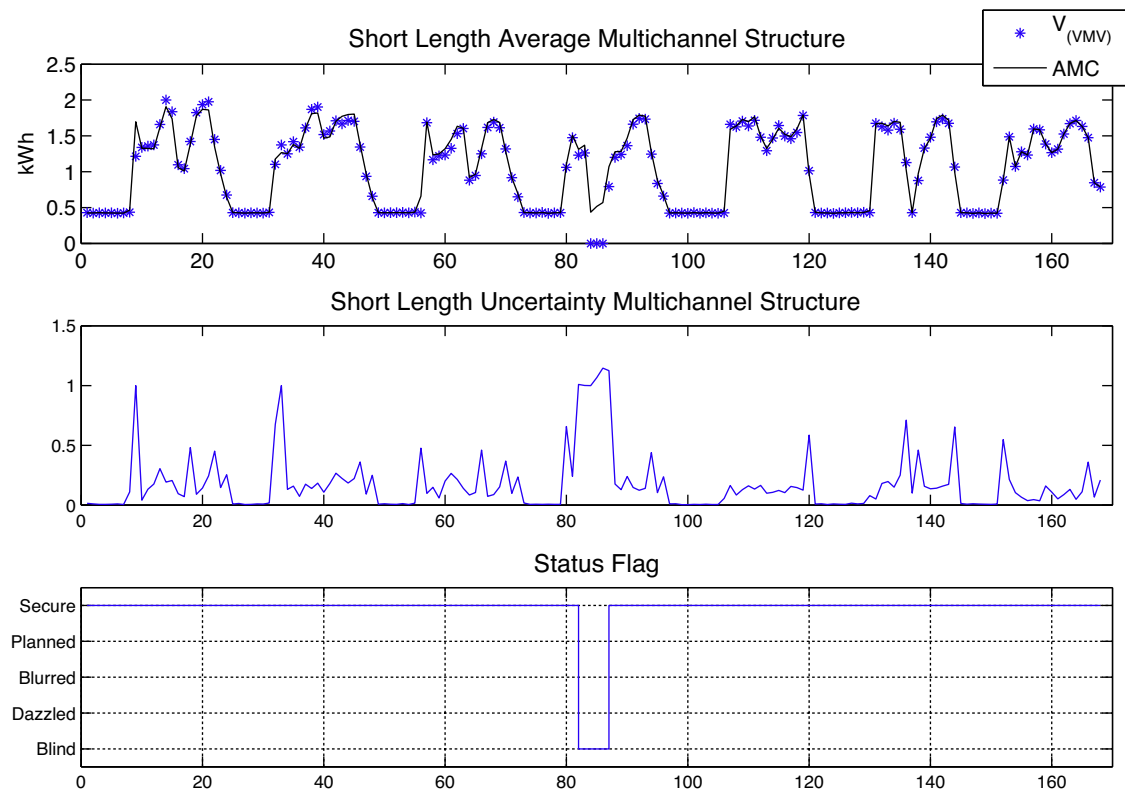


Fig. 16. Short length multichannels and status flag – week 8 of simulated data.

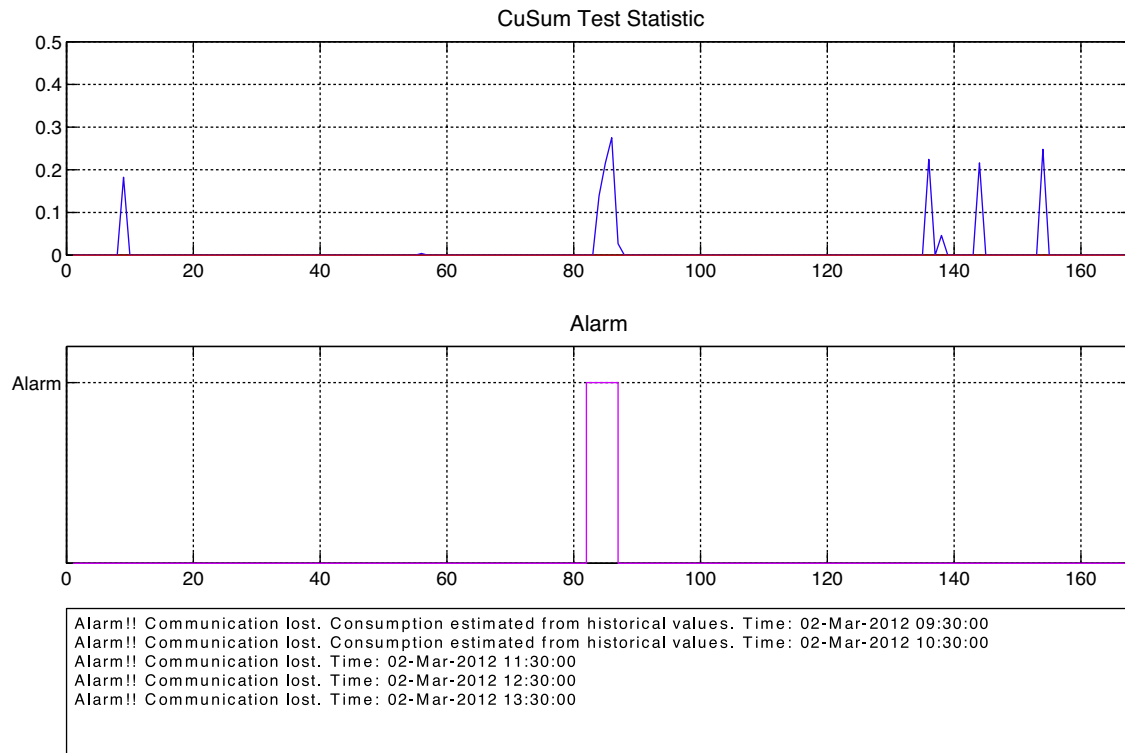


Fig. 17. CuSum statistic and alarm – week 8 of simulated data.

Two situations of unusual consumption occur during monitoring: in the third week of monitoring unusual consumption occurs in Sunday due to the application of an exam in the building and in the fifth week of monitoring a holiday occurs on Friday. As both situations are known to occur, status flag is set as Planned during the occurrence of these events.

Fig. 20 shows the Short Length Average Multichannel Structure, the Short Length Uncertainty Multichannel Structure and the Status flag for the third week of monitoring. Note that the status flag changes to planned during the planned event and a minimal effect of this unusual consumption is observed in the average and uncertainty structures since the smoothing factor used to compute this values were set to a minimal value. No alarm is generated.

Fig. 10 shows the Short Length Average Multichannel Structure, the Short Length Uncertainty Multichannel Structure and the Status flag for the fifth week of monitoring. The status flag changes to planned during the holiday and again a minimal effect is shown. However the holiday also influences on Thursday's energy consumption which is lower than typical. Since this behavior was not planned this unusual consumption has a greater influence on the computed values for the average and uncertainty.

Fig. 21

Fig. 22 shows the occurrence of alarm on Thursday due to the lower energy consumption.

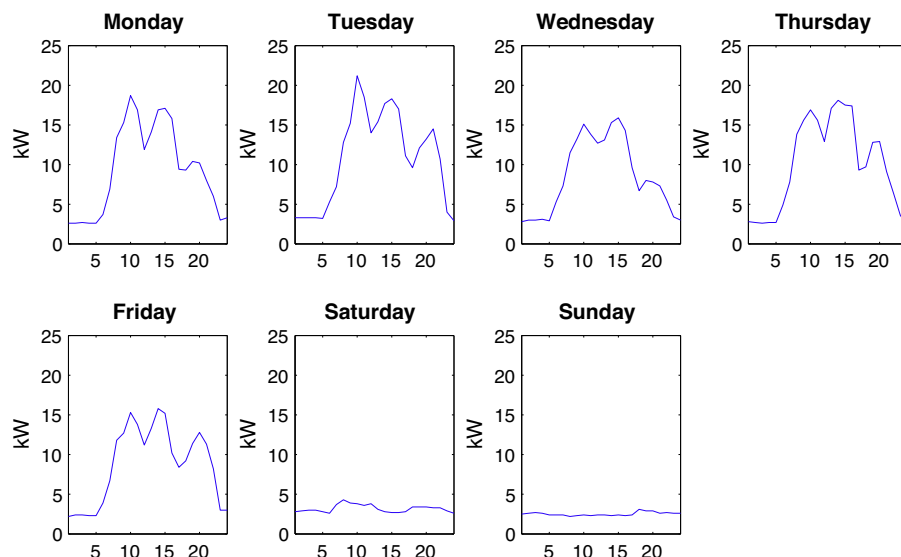


Fig. 18. Daily energy consumption of the Brazilian Educational Complex.

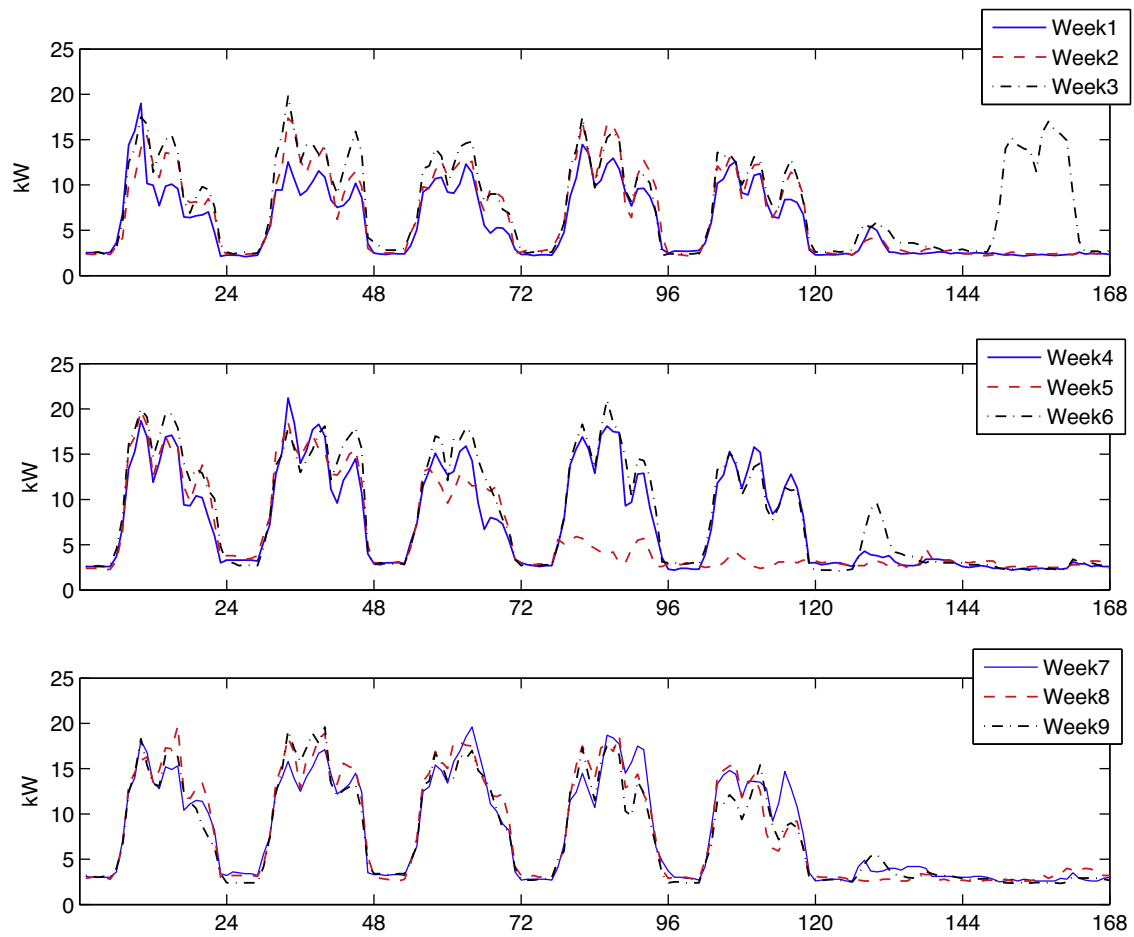


Fig. 19. Building monitored energy consumption of the Brazilian Educational Complex.

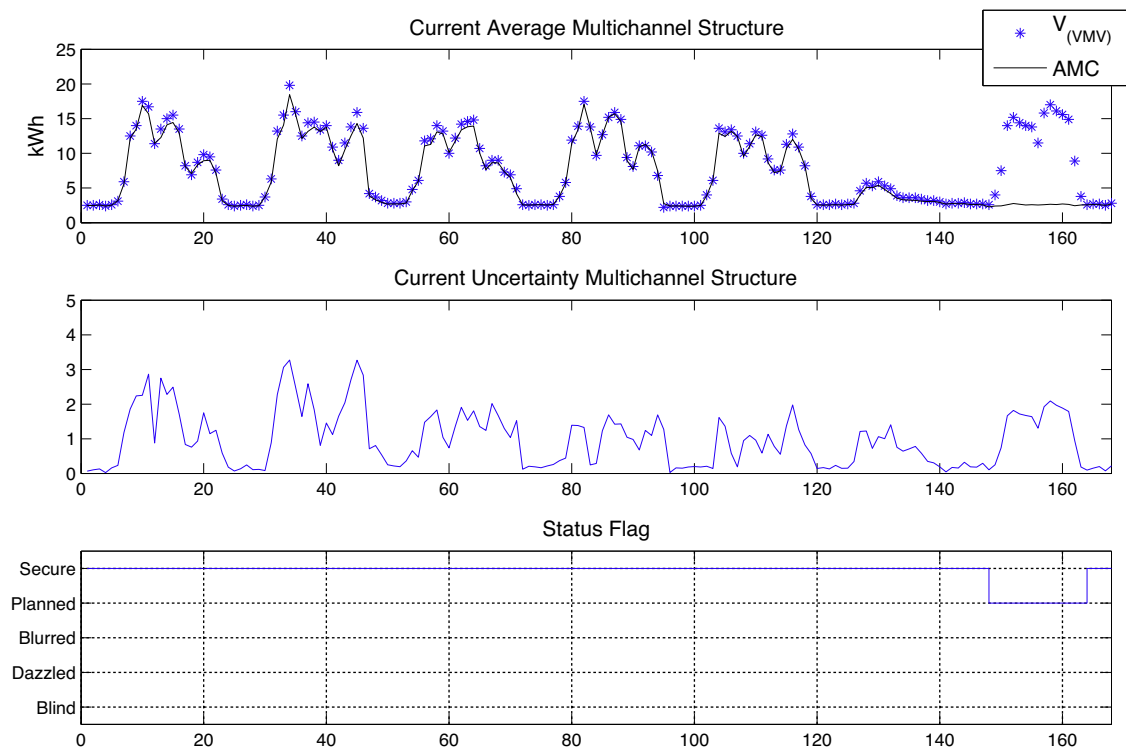


Fig. 20. Short length multichannels and status flag – week 3 of monitoring.

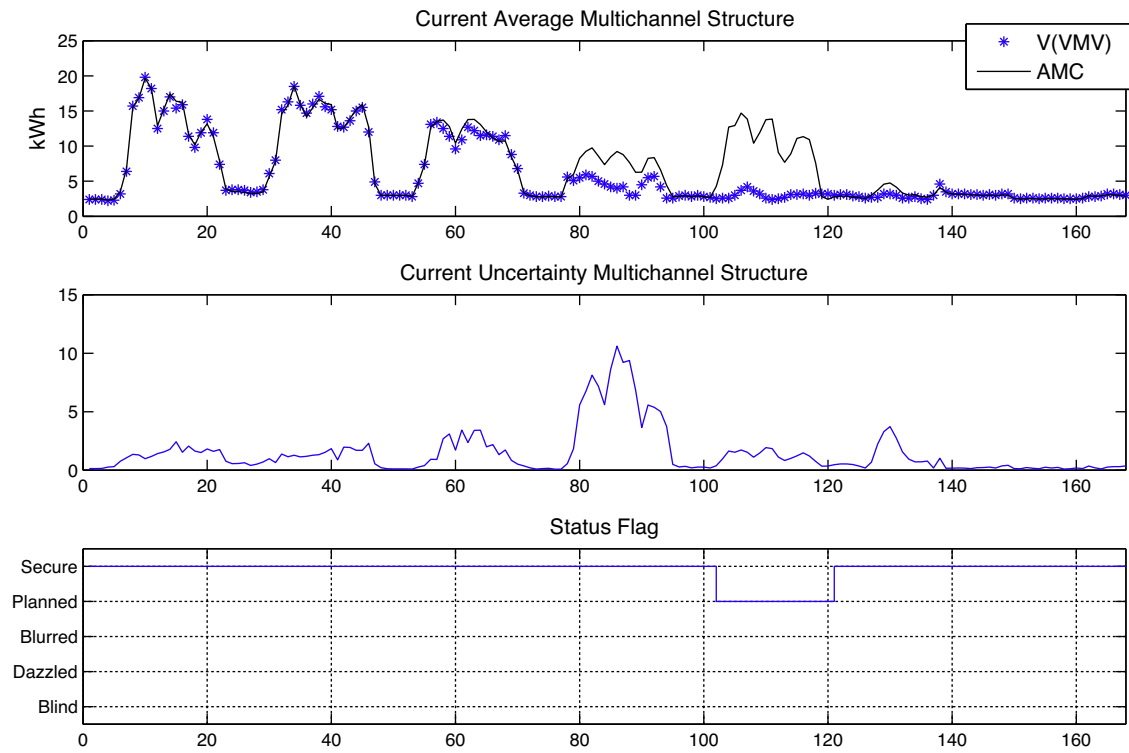


Fig. 21. Short length multichannels and status flag – week 5 of monitoring.

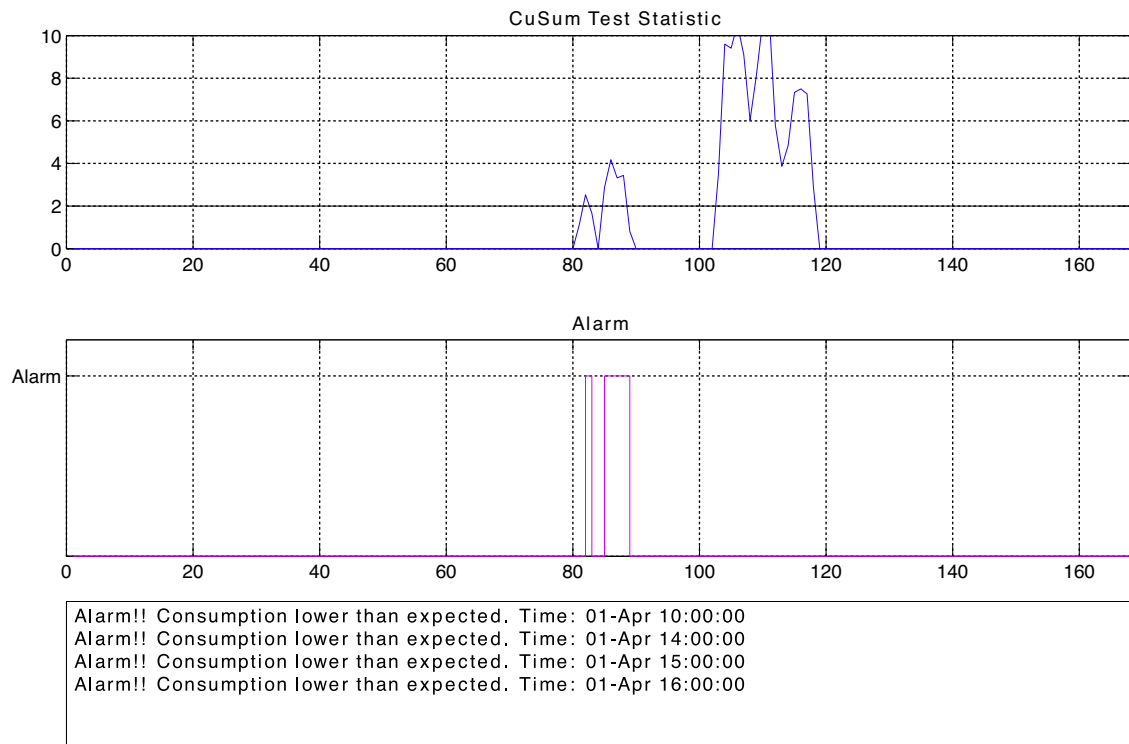


Fig. 22. CuSum statistic and alarm – week 5 of monitoring.

5. Conclusions

The methodology presented illustrated a feasible just in time and systematic practice proposal for monitoring, assessment and tracking of the operating conditions of a building. An SPC scheme was shown to be appropriate for tracking, analyzing and reporting energy consumption data, determining if this consumption is under statistical process control. The strategy presented used a modified Multichannel Structure easily designed and tuned using algorithms suited for implementation of MTS modules using low cost microcontrollers.

The statistics pattern of a building's energy consumption during a predetermined cycle was modeled and estimated using the Multichannel Structure. The amount of data and its computational requirements and cost were shown to be dramatically reduced, resulting in an information compression scheme. Besides, the monitoring system was shown to be an efficacious scheme for the application of SPC techniques and its metrics using an automated statistical decision process for signaling alarms or reporting interpreted faults of unpredicted or unusual behavior of current energy demand of a building.

Simulated experiments using the software Energy Plus was shown to be a valuable tool for designing and tuning the MTS presented. Experimental results illustrated the value of the proposed methodology for the challenging task of just in time characterization and monitoring of energy usage.

Results obtained by the case study evidenced that modeling the statistical behavior of energy consumption data into Multichannel Structures is suitable for monitoring using statistical process control techniques. The main challenging to get good performance in using this technique is to model the variable from reliable data which represent the system operating in statistical process control, stable and predictable conditions.

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